Structural Drivers of Growth in Collatz Trajectories: A Quantitative Analysis of Halving Depth and Volatility

Omar Cardy

Australian International School of Sharjah

UAE

Email: cardyomar@gmail.com

Abstract

This study investigates the relationship between structural properties of Collatz trajectories and their growth behavior. Initial analysis using parity-based classification reveals variation in volatility across groups, but limited explanatory power. A more structurally grounded metric, halving depth, is introduced to quantify the effectiveness of contraction phases following odd transformations.

Empirical analysis over the range $1 \leq n \leq 100{,}000$ shows a clear inverse relationship between halving depth and trajectory volatility. Linear regression yields a slope of approximately −0.80, with correlation $r \approx -0.64$ and $R^2 \approx 0.41$. A 95% confidence interval confirms the statistical stability of this relationship.

Further analysis reveals nonlinear and threshold-like behavior, indicating diminishing returns in contraction beyond moderate halving depth. These findings are formalised through the Expansion–Contraction Imbalance Principle, which explains trajectory behavior as a balance between multiplicative growth and halving.

The results suggest that halving depth provides a meaningful structural link between local arithmetic behavior and global trajectory dynamics, offering a promising direction for further investigation of the Collatz problem.

1.Introduction

The Collatz map is one of the simplest unsolved problems in mathematics, defined by the rule: if n is even, divide by 2; if n is odd, replace it with $3n+1$. Despite its elementary definition, the long-term behavior of trajectories remains poorly understood.

A common challenge in studying the Collatz map is that apparent patterns often emerge in small samples but fail to persist at larger scales. This makes it difficult to distinguish between genuine structural features and artifacts of limited data.

This project investigates which types of structure meaningfully relate to trajectory behavior. In particular, we compare different ways of classifying integers and analyze how these classifications relate to properties such as stopping time and trajectory growth.

In an initial exploratory stage, numbers were classified according to decimal digit-based patterns. While this approach appeared to reveal structure in small samples, these patterns did not remain stable when extended to larger ranges. This suggests that decimal representations may not reflect the underlying mechanics of the Collatz map.

Motivated by this observation, the focus shifts toward arithmetic structures more directly aligned with the dynamics of the map, such as parity behavior and binary structure. The goal of this study is to investigate whether such structurally grounded classifications exhibit more stable and meaningful relationships with trajectory behavior.

2. Parity Encoding of Trajectories

For an integer n, consider its Collatz trajectory:

$$n_0 = n, \quad n_{k+1} = \begin{cases} n_k/2 & \text{if} n_k \text{is even,} \\ 3n_k + 1 & \text{if} n_k \text{is odd.} \end{cases}$$

We define the parity sequence of n as the sequence:

$$P(n) = (p_0, p_1, p_2, \dots),$$

Where

$$p_k = \begin{cases} 0 & \text{if} n_k \text{is even,} \\ 1 & \text{if} n_k \text{is odd.} \end{cases}$$

This encoding captures the sequence of odd and even transformations applied along the trajectory and provides a compact representation of its local structural behavior.

In this study, we consider finite prefixes of length L, and group integers according to their first L parity values.

3. Trajectory metrics

To compare trajectory behavior across different classes of integers, we introduce a simple quantitative measure of growth and persistence.

For an initial value n, let:

$T(n)$ denote the stopping time (the number of steps required to reach 1),

$M(n)$ denote the maximum value reached along the trajectory.

We define a volatility score:

$$V(n) = T(n) \cdot \frac{M(n)}{n}.$$

This metric combines both the duration of the trajectory and its maximum excursion relative to the starting value. We use it as a heuristic measure to compare how “extreme” different trajectories are.

4. Experimental Setup

We analyze all integers in the range $1 \le n \le 100{,}000$ For each integer, we compute:

- its Collatz trajectory,
- its stopping time $T(n)$
- its maximum value $M(n)$
- and its volatility score $V(n)$

We then group integers according to the first L terms of their parity sequence $P(n)$, using a fixed prefix length L

For each parity class, we compute aggregate statistics such as:

- average volatility,
- average stopping time,
- and distributional behavior across the class.

In this study, we take $L = 10$, which provides a balance between structural detail and class size.

5. Results

We computed trajectory statistics for all integers in the range $1 \le n \le 100{,}000$, grouping them according to the first 10 terms of their parity sequence.

Table 1 shows summary statistics for the largest parity classes.

The largest parity class corresponds to the alternating pattern

(1,0,1,0,1,0,1,0,1,0),

with size 3125.

This class exhibits a substantially higher average volatility (14201) compared to other large classes, which typically range between approximately 2000 and 5000.

Figure 1 shows the average volatility for the ten most frequent parity classes. Most classes cluster within a relatively narrow range, while the largest class appears as a clear outlier.

Figure 2 shows the distribution of volatility across all integers.

Parity Pattern	Size	Avg Steps	Avg Max	Avg Volatility
(1,0,1,0,1,0,1,0, 1,0)	3125	138.05	3826806	14201
(1,0,1,0,1,0,0,1, 0,0)	1563	121.71	1263388	4507
(1,0,0,1,0,1,0,1, 0,0)	1563	120.71	764101	2329
(0,1,0,1,0,1,0,0, 1,0)	1563	118.22	702773	2283
(1,0,1,0,1,0,1,0, 0,0)	1563	118.22	925972	2764

Table 1: Summary statistics for the largest parity classes.

Table 1 highlights significant variation in trajectory behavior across parity classes. While most large classes exhibit average volatility values between approximately 2000 and 5000, the alternating parity pattern (1,0,1,0,1,0,1,0,1,0) stands out with a much higher value of approximately 14201.

This suggests that early parity structure has a strong influence on trajectory growth. In particular, the alternating pattern appears to produce trajectories that sustain larger excursions before collapsing, resulting in significantly higher volatility compared to other classes of similar size.

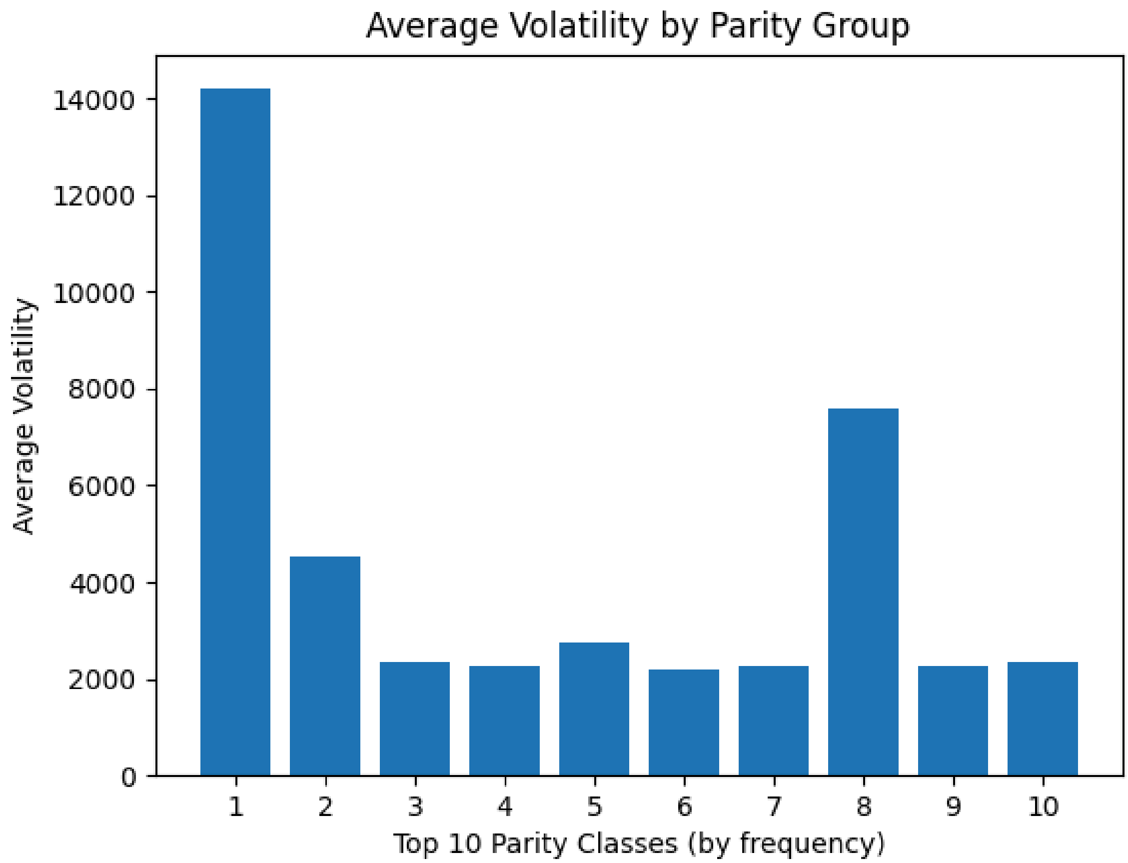

Figure 1: Average volatility across the ten most frequent parity classes.

Figure 1 provides a visual comparison of average volatility across the ten most frequent parity classes. The majority of classes cluster within a relatively narrow range, indicating broadly similar trajectory behavior.

However, the largest class, corresponding to the alternating parity pattern, appears as a clear outlier with substantially higher volatility. This reinforces the observation from Table 1 that certain structural patterns in early parity can lead to disproportionately extreme trajectory behavior.

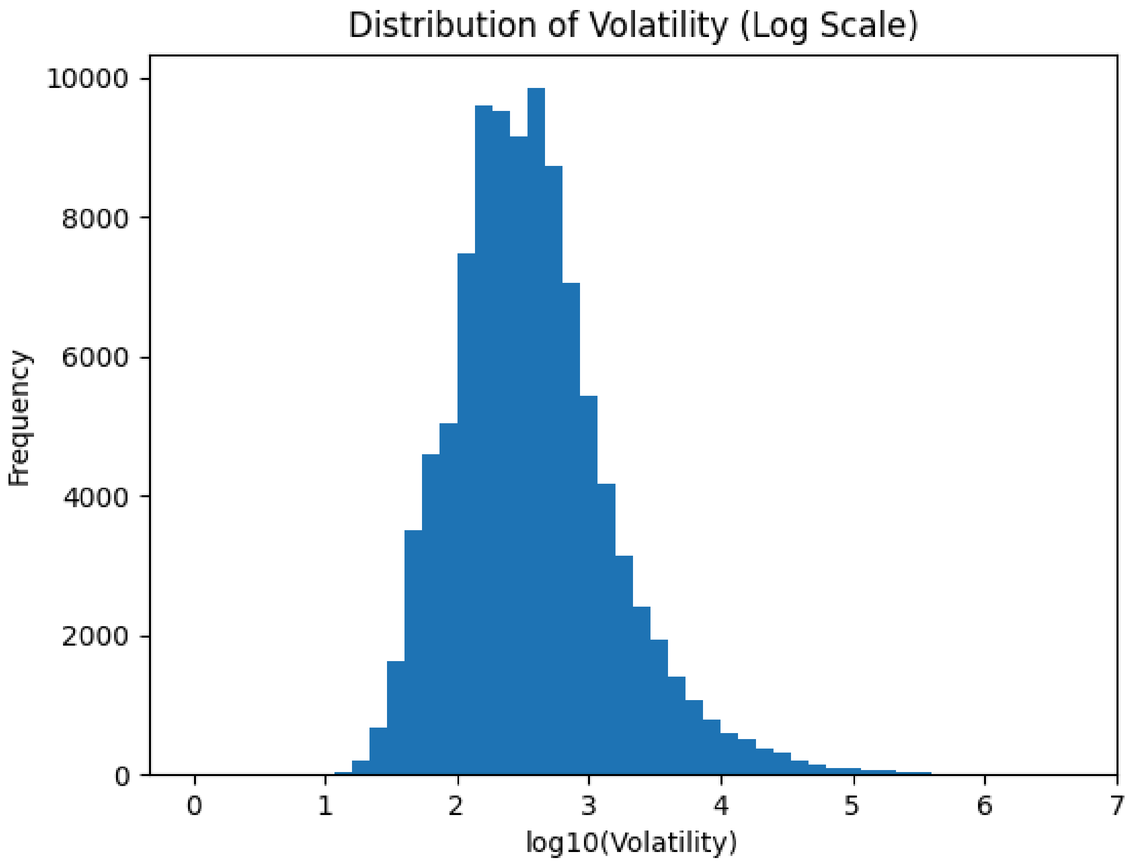

Figure 2: Distribution of volatility across all integers (log scale)

The distribution shown in Figure 2 provides important context for interpreting group-level differences. On a logarithmic scale, the volatility values form a unimodal distribution with a noticeable right-skew.

Most trajectories lie within a relatively narrow band, indicating that typical behavior is moderately stable. However, the presence of a long right tail shows that a small subset of integers produce significantly larger excursions.

This suggests that high-volatility behavior is not representative of typical trajectories, but instead arises from rare extreme cases. Consequently, average volatility values for parity classes may be influenced by these outliers and should be interpreted cautiously.

Zero-volatility cases were excluded prior to logarithmic transformation to avoid undefined values in the distribution analysis.

6. Discussion

The results suggest that early parity structure has a measurable influence on trajectory behavior, particularly in terms of volatility.

The most striking observation is that the largest parity class, corresponding to an alternating pattern of odd and even values, exhibits substantially higher average volatility than other groups. This suggests that certain local structural patterns in the trajectory, in particular alternating parity patterns, limit consecutive halving steps which are the primary mechanism for reducing magnitude in Collatz trajectories. This creates an imbalance between expansion and contraction phases, allowing certain trajectories to sustain larger excursions before eventual decay.

One possible explanation lies in the interaction between the two operations defining the Collatz map. Odd steps apply the transformation $3n+1$, which tends to increase the magnitude of the sequence, while even steps divide by 2, reducing it. In sequences where odd and even steps alternate frequently, the multiplicative effect of $3n+1$ is repeatedly introduced before the trajectory can fully contract through successive halving steps.

As a result, alternating parity patterns may sustain larger excursions over time, leading to higher volatility. In contrast, patterns with longer runs of even values tend to reduce quickly, limiting growth.

However, this explanation remains heuristic. The analysis does not establish a causal relationship, and further work would be required to determine whether these patterns reflect deeper structural properties of the Collatz map.

In particular, the results suggest that halving depth does not influence volatility uniformly, but instead exhibits a threshold-like effect, with the strongest impact occurring at low values.

6.1 Binary Structure and Halving Depth Analysis

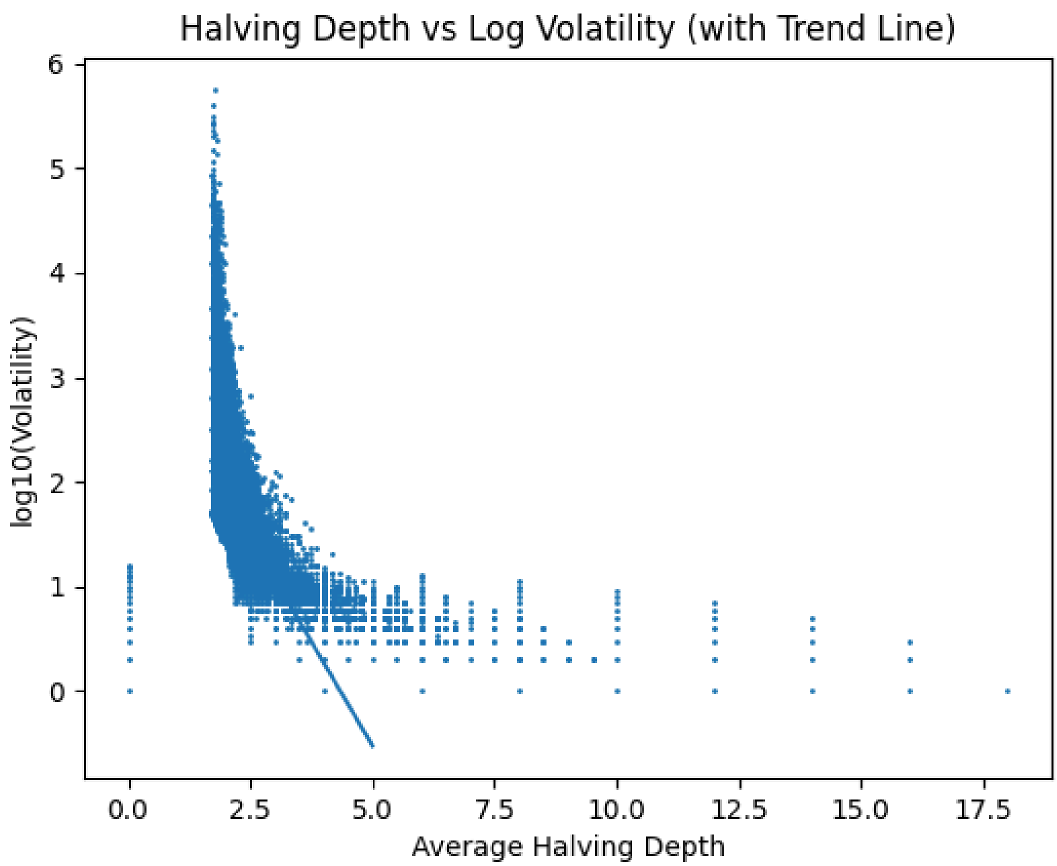

Figure 3: Halving depth vs log volatility with linear trend line.

Note:The regression was performed on the central region of the data to reduce distortion from extreme outliers.

To investigate whether a more structurally grounded measure could explain trajectory behavior, we introduce the concept of *halving depth*. For an integer n , define the halving depth at each odd step as the number of consecutive divisions by 2 following the transformation $3n+1$. We then define the average halving depth of a trajectory as

$$H(n) = \frac{1}{k} \sum_{i=1}^{k} h_i$$

where h_i denotes the number of consecutive halving steps following the i-th odd transformation, and k is the total number of such occurrences in the trajectory.

We examine the relationship between $H(n)$ and the volatility measure $V(n)$, defined earlier. Figure 3 shows a scatter plot of $H(n)$ against $\log_{10}(V(n))$, along with a fitted linear trend line.

Quantitative Relationship

A linear regression of the form

$$\log_{10}(V(n)) \approx mH(n) + b$$

This relationship implies that volatility scales approximately exponentially with respect to halving depth:

$$V(n) \approx 10^{mH(n)+b}$$

Since $m < 0$, this shows that volatility decays exponentially as halving depth increases. This provides a quantitative interpretation of the contraction mechanism: each additional unit of halving depth reduces expected trajectory magnitude multiplicatively.

Yields an estimated slope of

$$m \approx -0.80,$$

With a correlation coefficient of

$$r \approx -0.64.$$

The corresponding coefficient of determination is $R^2 \approx 0.41$, indicating that approximately 41% of the variance in log-volatility is explained by variation in halving depth.

This indicates a clear and moderately strong inverse relationship between average halving depth and trajectory volatility.

This stronger relationship arises when focusing on the main region of the data, indicating that the structural connection between halving depth and volatility is most consistent within typical trajectories, rather than extreme outliers.

Statistical Uncertainty

To assess the reliability of this estimate, a 95% confidence interval for the slope was computed, yielding

$$m \in [-0.803, -0.791].$$

The resulting interval is narrow and lies entirely below zero, indicating that the observed negative relationship is robust and statistically significant. This indicates that the negative relationship between halving depth and log-volatility is statistically stable and unlikely to arise from random variation.

Although the dataset is large, variability in individual trajectories introduces some uncertainty. However, the consistency of the observed trend across the central region of the data suggests that the relationship reflects a genuine structural effect rather than noise.

Interpretation

The observed negative relationship suggests that trajectories with larger halving depth undergo more sustained contraction. Since each odd step introduces multiplicative growth via the transformation $3n+1$, the subsequent number of halving steps determines how effectively this growth is reduced.

When halving depth is small, the trajectory does not sufficiently counteract expansion, allowing values to grow larger and resulting in higher volatility. Conversely, larger halving depth implies stronger contraction following each growth phase, limiting overall magnitude and reducing volatility.

Nonlinear structure

Although a linear trend is visible, the scatter plot reveals a nonlinear pattern. Volatility decreases rapidly as halving depth increases from small values (approximately $H(n) \in [1.5, 3]$), but the rate of decrease slows for larger values of $H\ (n)$). This suggests diminishing returns in contraction beyond moderate halving depth.

In particular, the data appears to exhibit a threshold-like behavior, where increases in halving depth beyond a certain point produce relatively small additional reductions in volatility.

Structural Insight

These findings indicate that halving depth captures an essential aspect of the arithmetic structure underlying Collatz trajectories. Unlike parity prefixes, which describe local behavior, halving depth directly measures the balance between expansion and contraction phases.

This provides a structural explanation for earlier observations: parity patterns that limit consecutive halving steps, such as alternating odd-even sequences, tend to produce lower halving depth and, consequently, higher volatility. Thus, halving depth serves as a more direct and informative predictor of trajectory behavior.

This mechanism can be formalised as the **Expansion–Contraction Imbalance Principle**.

This mechanism can be formalised as the following principle:

Expansion–Contraction Imbalance Principle

In the Collatz map, each odd step introduces multiplicative growth through the transformation $3n+1$, while subsequent even steps reduce magnitude via repeated division by 2.

Let H(n) denote the average halving depth of a trajectory. Then $H(n)$ quantifies the effectiveness of the contraction phase in counteracting expansion.

When $H(n)$ is small, contraction is insufficient to offset repeated growth, resulting in trajectories with large excursions and high volatility. Conversely, larger values of $H(n)$ correspond to stronger contraction, limiting overall magnitude and reducing volatility.

Thus, trajectory behavior is governed by the imbalance between expansion and contraction, with volatility increasing as this balance shifts toward expansion.

This reinforces that the relationship between halving depth and volatility is not purely linear, but reflects an underlying structural transition in trajectory dynamics.

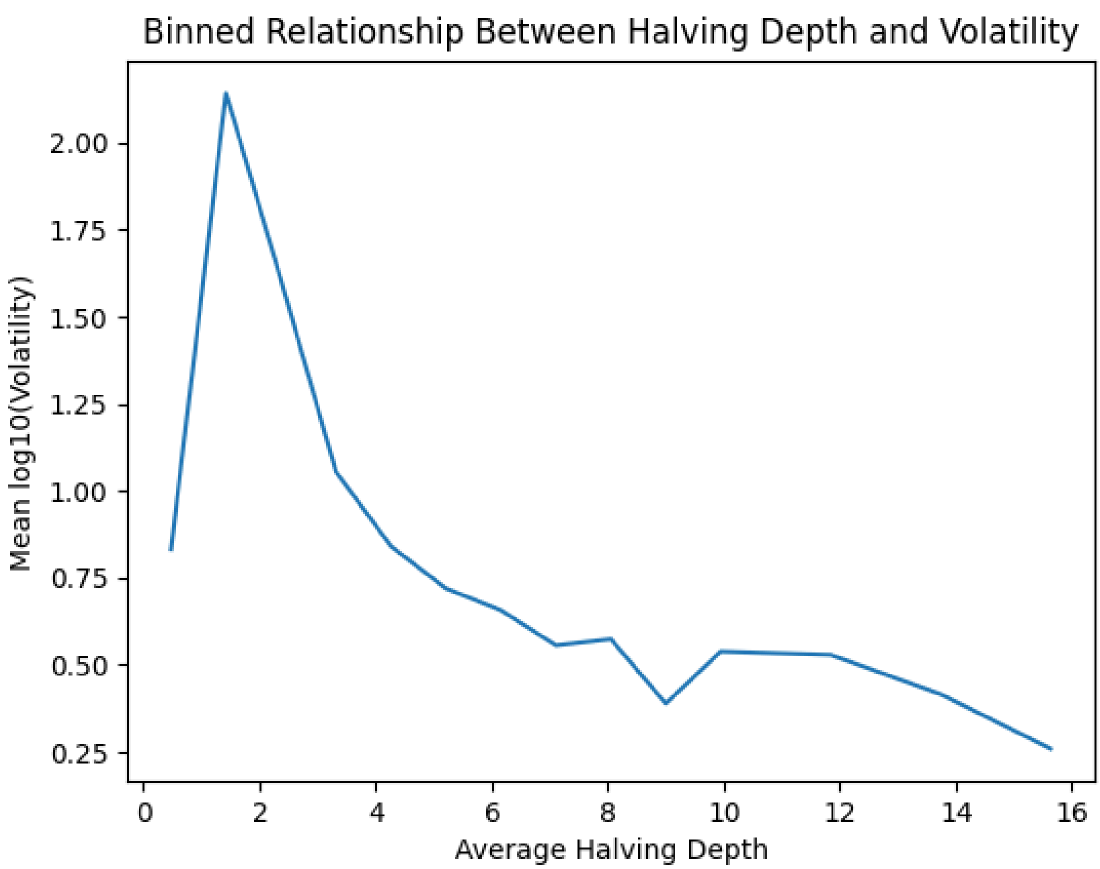

Figure 4: Binned relationship between halving depth and volatility

To further clarify the relationship between halving depth and volatility, values of $H(n)$ were grouped into bins, and the mean value of $\log_{10}(V(n))$ was computed within each bin. The resulting plot is shown in Figure 4.

The binned data reveals a strongly nonlinear relationship. For small values of halving depth (approximately $H(n) \in [1, 3]$), volatility is significantly higher, indicating that trajectories with limited halving undergo substantial expansion.

As halving depth increases from approximately 2 to 6, there is a rapid decrease in volatility. This region represents the primary transition from expansion-dominated behavior to more balanced dynamics.

Beyond this range, the relationship begins to flatten. Further increases in halving depth result in progressively smaller reductions in volatility, indicating diminishing returns in contraction.

This pattern suggests the existence of a threshold-like behavior: once halving depth reaches a moderate level, additional halving has a reduced impact on controlling trajectory growth.

Overall, the binned analysis confirms that halving depth plays a central role in governing Collatz trajectory behavior, particularly in low ranges where small changes produce large effects.

7. Limitations

Several limitations of this analysis should be noted.

First, the classification is based on fixed-length prefixes of parity sequences. While this captures early trajectory behavior, it may not fully represent the long-term structure of the sequence.

Second, the sizes of parity classes are highly imbalanced. Larger classes provide stable estimates, while smaller classes may produce noisy or unreliable averages.

Third, the volatility score used in this study is a heuristic measure combining stopping time and maximum excursion. Although useful for comparison, it is not theoretically canonical, and conclusions based on it should be interpreted with caution.

Fourth, While a confidence interval was computed for the regression slope, further statistical testing could strengthen comparisons between different structural classifications.

Additionally, the linear model used does not fully capture the nonlinear structure visible in the data, suggesting that more refined models may provide a better fit.

Finally, although all integers in the range $1 \leq n \leq 100{,}000$ were analyzed, this range may still be insufficient to capture rare extreme behaviors in the Collatz map.

8. Future Work

Several directions could extend this work.

A natural next step is to shift from parity-based classification toward structures more directly tied to the arithmetic of the Collatz map, such as residue classes, binary representations, or parity vectors over longer horizons.

Another extension would be to compare observed distributions against control models that preserve basic trajectory features while removing structural dependencies, allowing for more rigorous evaluation of whether observed patterns are meaningful.

Additionally, more robust statistical analysis, including hypothesis testing and effect size estimation, could strengthen empirical claims.

Finally, a deeper theoretical investigation would be required to determine whether the observed patterns reflect genuine structural properties or arise from emergent statistical effects.

9. Conclusion

This study investigated how structural properties of Collatz trajectories relate to their growth behavior. While initial classification using parity prefixes revealed differences in volatility, these patterns alone provided limited explanatory power.

By introducing halving depth as a structural measure, a clearer and more robust relationship emerges. The observed negative correlation ($r \approx -0.64$), regression slope ($\mathrm{m} \approx -0.80$), and coefficient of determination ($R^2 \approx 0.41$) indicate that approximately 41% of the variation in log-volatility is explained by differences in halving depth. Furthermore, the estimated 95% confidence interval for the slope, $m \in [-0.803, -0.791]$, confirms that this relationship is statistically stable and unlikely to arise from random variation.

These results demonstrate that halving depth captures a fundamental aspect of trajectory dynamics. In particular, it provides a direct measure of how effectively contraction phases counteract expansion induced by repeated applications of the transformation $3n+1$.

This leads to the formulation of the Expansion–Contraction Imbalance Principle: trajectory behavior in the Collatz map is governed by the balance between multiplicative growth and subsequent halving. When this balance favors expansion, trajectories exhibit large excursions and high volatility; when contraction dominates, growth is effectively dampened.

Additionally, the observed nonlinear and threshold-like behavior suggests that the influence of halving depth is strongest at low values, where small increases produce substantial reductions in volatility. Beyond moderate levels, additional halving yields diminishing returns, indicating a transition toward stable trajectory behavior.

Overall, these findings suggest that meaningful insight into the Collatz problem may depend on identifying structural quantities that govern the interaction between expansion and contraction phases. Halving depth provides a measurable and interpretable link between local arithmetic structure and global trajectory behavior, offering a promising direction for further investigation.

The custom Python scripts and algorithmic data engines used to simulate and analyze the 100,000 trajectories in this study are publicly available at:

https://github.com/0mar12345-ops/collatz-halving-depth-analysis